

TENNIS BALL CATAPULT

Description: Students will build and calibrate their own free-standing (not handheld) trajectory device that must be capable of “lobbing” a tennis ball at a target placed between 5 and 8 meters away. Students MUST bring and wear impact-resistant safety goggles. **This is a PRE-BUILT EVENT.**

Number of Participants: 2

Impound: YES

The Competition:

1. Each team will bring their own homemade catapult or trebuchet and their own standard unaltered regulation tennis balls to the competition. It is suggested to bring three tennis balls. Print your team number prominently on the catapult and balls. If the team fails to bring their own tennis balls, they may be supplied by the judge at a penalty of +20 points.
2. Launch force must be provided by gravity or elastic solids (such as springs, rubber bands, bungee cords, etc.).
3. The device may not be more than 50 cm above the ground before, during or after launching. The total weight of the device must be under 50 pounds. Students must be able to move their device from the impound area to the launch site without adult assistance. If the device violates these design parameters, the team will be placed in Tier 2
4. Each device should be designed and built by the students (adult construction assistance is OK). Each device should be designed to operate safely at all times. No points will be awarded for design.
5. The device must sit upon rubber or soft material “feet” to prevent floor damage when moving and launching.
6. The device will sit on a level area of ground and fire at the target area that will also be at ground level. The target area will be a freshly raked sand pile of at least 1 meter. A small object will mark the target.
7. The event may take place outside.
8. **The team may not use any digital or electronic devices to aid in aiming or measuring.**
9. Prior to the competition, teams should prepare a design log. The design log will be submitted during impound and should include:
 - a. At least 2 diagrams showing different views of the device and details of construction
 - b. A graph or data chart compiled from launch practice, graph title, labeled x- and y- axes and using appropriate units for usage of their device.
 - c. The log should be clearly labeled with the team number.
 - d. Students should be prepared to answer questions about their device and graph.

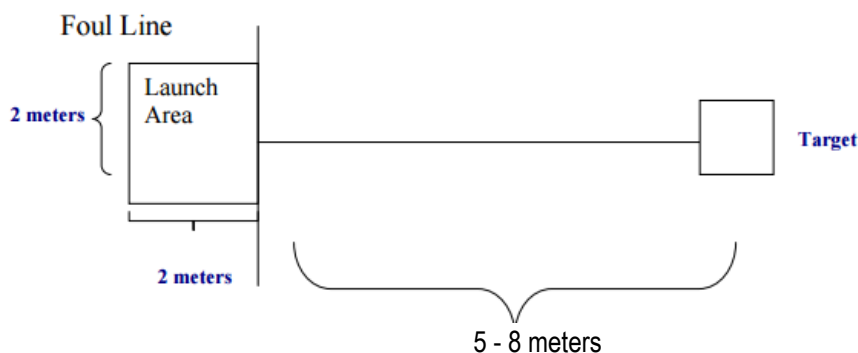
Impound:

1. During the designated impound period, the team must impound their catapult/trebuchet, any materials needed to alter the device during the testing phase, a tennis ball, impact resistant safety goggles, and their design log. All items should be clearly marked with the team number.
2. During impound, Event Supervisors will evaluate all devices for safety and determine if the device is suitable for testing and meet the design requirements.

Launching:

1. Once the team is called to set up for launching, no coaching or support from outside is allowed. This could result in the team being placed in Tier 3.

2. Impact-resistant safety goggles (or glasses) are required during launch time. If team members do not wear safety goggles while they are setting up and running their device, they will be given an initial warning and a penalty of +20 points. If a second warning is given, the team will immediately be placed in Tier 2.
3. Once teams enter the event area to compete, they may not leave the area or receive outside assistance, materials or communications until they are finished competing. A violation of this rule will result in the team being placed in Tier 3.
4. The distance from the “foul line” to the center of the target will be announced at the start of the competition. Students may place their catapult at any point behind the foul line up to 2 meters. See image below.



5. Total time once the team is called is 5 minutes. Each team will have a one-shot practice round after which they will shoot at the target two times. The second and third shots will count toward the official score.
6. The practice shot measurement will only be used in the case of a tie.
7. Teams must give ample warning to the supervisors prior to launch. Warning may be “3,2,1 launching...” etc. A penalty of +20 points will be assessed for failure to warn supervisors before each launch.
- 8.
9. Only successful launches will be measured. Launches are successful when a Tennis Ball's first bounce goes beyond the foul line. Only 2 shots are allowed, beyond the practice shot, regardless if they are successful or not. (An unsuccessful launch will be counted as 2m (200 cm) from the target for the purposes of averaging launches.)
10. The device may be moved by the participants within the 2-meter square launch area after each attempted launch. Teams may make adjustments to their device, between shots, while in the launch area using only materials that were impounded with the device. They may not change the device in any way that would violate impound rules or design parameters..
11. The distance from the target will be measured after each shot, including the practice shot, and announced to the team to allow them to make adjustments based on their data chart. **All measurements will be announced in centimeters.**
12. The distance from the center of the target will be measured to the tennis ball's initial point of impact as determined by the supervisor. Tennis balls landing outside of the sand pile area, less than 2 meters from the launch area, will also be measured according to the initial point of impact as determined by the supervisor. Tennis balls hitting the target on initial impact will receive a score of zero. Tennis balls with an initial impact beyond 2 meters will be scored as 2 meters.
13. No tennis balls should be shot before or during the catapult event except during the 5-minute specified practice round or actual competition.
14. Teams will be allowed to use their graph or any information from the design log during competition.
15. The event supervisor will make every attempt to advise students of any penalties incurred during the launch.

Scoring:

1. Low score wins.
2. Score is the average of the distance from the target in centimeters, measured to a single centimeter precision. 1 centimeter is 1 point
3. Devices which do not meet the design requirements will result in the team being placed in Tier 2.
4. Penalties may be assessed as listed in the rules.
5. Ties will be broken by the distance from the target on the third tiebreaker launch.

Tennis Ball Catapult Scoring Rubric

Team #: _____

School/Team Name: _____

		Does the Catapult meet the requirements? (Y/N)
IMPOUND :		
Catapult determined to be safe?	N= Cannot compete;	
Catapult sits upon rubber or soft material so as to not damage floor? Cannot compete	N =	
Catapult is within design parameters. Less than 50 cm tall and less than 50 ponds	N = Tier 2	
Catapult labeled with team #?	N= -Tier 2	
Tennis ball(s) impounded and labeled	N= Tier 2	
Design Log provided?	N = Tier 2	
Impound on time?	N = Tier 2	
	TIER?	

LAUNCH SCORE :

Points

Practice Launch - will be measured and announced to team and used only for breaking ties. (enter below)	
Launch #1 distance from target _____ cm failed launch = 200 cm	
Launch #2 distance from target _____ cm failed launch = 200 cm	
Average of the two Launches=	

LAUNCH PENALTIES :

Penalty

Failure to announce launch (3,2,1 ...launching) + 20 points	
Student or device crosses launch line? +20 points	
Tennis Ball supplied by Event Supervisor +20 points	
Goggles violation +20 points or Tier 2	
Outside interference? Y= Tier 3	
TOTAL=	

PLEASE NOTE THAT THIS SCORESHEET IS A SAMPLE AND GUIDE- IN CASES OF CONFLICT WITH THE RULES, THE RULES TAKE PRECEDENCE OVER THE SCORESHEET.